



Current status and methods

- ★ Tau neutrino helicity

- from angular and kinematic correlations in polarized τ -decays, for example:

$$\tau^{\mp} \rightarrow \pi^{\mp} \pi^0 \nu_{\tau}.$$

CLEO found

$$h_{\nu_{\tau}} = -0.995 \pm 0.010 \pm 0.003.$$

Exercise 1

Using $E_\nu \approx 1 \text{ MeV}$, estimate the recoil energy and recoil velocity of B^*

Then, assuming the recoil loses directional memory after traveling a few angstroms in a solid, estimate τ_{stop}